

MARK SCHEME for the May/June 2008 question paper

9702 PHYSICS

9702/04

Paper 4 (A2 Structured Questions), maximum raw mark 100

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Section A

- 1 (a) (i) angle (subtended) at centre of circle
by an arc equal in length to the radius (of the circle) B1
B1 [2]
- (ii) angle swept out per unit time / rate of change of angle
by the string M1
A1 [2]
- (b) friction provides / equals the centripetal force B1
 $0.72 W = m d \omega^2$ C1
 $0.72 mg = m \times 0.35 \omega^2$
 $\omega = 4.49 \text{ (rad s}^{-1}\text{)}$ C1
 $n = (\omega / 2\pi) \times 60$ B1
 $= 43 \text{ min}^{-1} \text{ (allow 42)}$ A1 [5]
- (c) either centripetal force increases as r increases
or centripetal force larger at edge M1
so flies off at edge first A1 [2]
($F = m r \omega^2$ so edge first – treat as special case and allow one mark)
- 2 (a) molecule(s) rebound from wall of vessel / hits walls B1
change in momentum gives rise to impulse / force B1
either (many impulses) averaged to give constant force / pressure
or the molecules are in random motion B1 [3]
- (b) (i) $p = \frac{1}{3} \rho \langle c^2 \rangle$ C1
 $1.02 \times 10^5 = \frac{1}{3} \times 0.900 \times \langle c^2 \rangle$
 $\langle c^2 \rangle = 3.4 \times 10^5$ C1
 $c_{\text{RMS}} = 580 \text{ m s}^{-1}$ A1 [3]
- (ii) either $\langle c^2 \rangle \propto T$ or $\langle c^2 \rangle = 2 \times 3.4 \times 10^5$ C1
 $c_{\text{RMS}} = 830 \text{ m s}^{-1} \text{ (allow 820)}$ A1 [2]
- (c) c_{RMS} depends on temperature (alone) B1
so no effect B1 [2]

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- 3 (a) (i) amplitude = 0.5 cm A1 [1]
- (ii) period = 0.8 s A1 [1]
- (b) (i) $\omega = 2\pi / T$
 $= 7.85 \text{ rad s}^{-1}$
correct use of $v = \omega \sqrt{(x_0^2 - x^2)}$ B1
 $= 7.85 \times \sqrt{\{0.5 \times 10^{-2}\}^2 - \{0.2 \times 10^{-2}\}^2}$
 $= 3.6 \text{ cm s}^{-1}$ A1 [3]
(if tangent drawn or clearly implied (B1)
 $3.6 \pm 0.3 \text{ cm s}^{-1}$ (A2)
but allow 1 mark for $> \pm 0.3$ but $\leq \pm 0.6 \text{ cm s}^{-1}$)
- (ii) $d = 15.8 \text{ cm}$ A1 [1]
- (c) (i) (continuous) loss of energy / reduction in amplitude (from the oscillating system) B1
caused by force acting in opposite direction to the motion / friction / viscous forces B1 [2]
- (ii) same period / small increase in period B1
line displacement always less than that on Fig.3.2 (*ignore first T/4*) M1
peak progressively smaller A1 [3]
- 4 (a) work done moving unit positive charge from infinity to the point M1
A1 [2]
- (b) (i) $x = 18 \text{ cm}$ A1 [1]
- (ii) $V_A + V_B = 0$ C1
 $(3.6 \times 10^{-9}) / (4\pi\epsilon_0 \times 18 \times 10^{-2}) + q / (4\pi\epsilon_0 \times 12 \times 10^{-2}) = 0$ C1
 $q = -2.4 \times 10^{-9} \text{ C}$ A1 [3]
(use of $V_A = V_B$ giving $2.4 \times 10^{-9} \text{ C}$ scores one mark)
- (c) field strength = (–) gradient of graph B1
force = charge $\times$ gradient / field strength *or* force $\propto$ gradient B1
force largest at $x = 27 \text{ cm}$ B1 [3]
- 5 (a) at $t = 1.0 \text{ s}$, $V = 2.5 \text{ V}$ C1
energy = $\frac{1}{2}CV^2$ C1
 $0.13 = \frac{1}{2} \times C \times (8.0^2 - 2.5^2)$ M1
 $C = 4500 \mu\text{F}$ A0 [3]
- (b) use of two capacitors in series in all branches of combination M1
connected into correct parallel arrangement A1 [2]

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- 6 (a) parallel (to the field) B1 [1]
- (b) (i) torque = $F \times d$
 $2.1 \times 10^{-3} = F \times 2.8 \times 10^{-2}$
 $F = 0.075 \text{ N}$
(use of 4.5 cm scores no marks) C1
A1 [2]
- (ii) zero A1 [1]
- (c) $F = BILN(\sin\theta)$ C1
 $0.075 = B \times 0.170 \times 4.5 \times 10^{-2} \times 140$ M1
 $B = 7.0 \times 10^{-2} \text{ T} = 70 \text{ mT}$ A0 [2]
- (d) (i) (induced) e.m.f. is proportional to / equal to rate of change of (magnetic) flux (linkage) M1
A1 [2]
- (ii) change in flux linkage = BAN
 $= 0.070 \times 4.5 \times 10^{-2} \times 2.8 \times 10^{-2} \times 140$ C1
 $= 0.0123 \text{ Wb turns}$
induced e.m.f = $0.0123 / 0.14$ C1
 $= 88 \text{ mV}$ A1 [3]
(Note: This is a simplified treatment. A full treatment would involve the averaging of $B \cos\theta$ leading to a $\sqrt{2}$ factor)
- 7 (a) charge is quantised / discrete quantities B1 [1]
- (b) (i) parallel so that the electric field is uniform / constant B1
horizontal so that *either* oil drop will not drift sideways
or field is vertical
or electric force is equal to weight B1 [2]
- (ii) $qE = mg$ C1
 $q \times 850 / (5.4 \times 10^{-3}) = 7.7 \times 10^{-15} \times 9.8$ C1
 $q = 4.8 \times 10^{-19} \text{ C}$ and is negative A1 [3]
- (c) charge changes by $1.6 \times 10^{-19} \text{ C}$ between droplets / integral multiples M1
so charge on electron is $1.6 \times 10^{-19} \text{ C}$ A0 [1]
- 8 (a) since momentum before combining is zero B1
momenta must be equal and opposite after B1
equal momenta so photon energies equal B1 [3]
- (b) $E = mc^2$ C1
 $= 9.1 \times 10^{-31} \times (3.0 \times 10^8)^2$
 $= 8.19 \times 10^{-14} \text{ (J)}$ C1
 $= (8.19 \times 10^{-14}) / (1.6 \times 10^{-13})$
 $= 0.51 \text{ MeV}$ A1 [3]

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Section B

- 9 (a) blocks labelled sensing device / sensor / transducer
processor / processing unit / signal conditioning B1 B1 [2]
- (b) (i) two LEDs with opposite polarities (*ignore any series resistors*)
correctly identified as red and green M1 A1 [2]
- (ii) correct polarity for diode to conduct identified
hence red LED conducts when input (+)ve or *vice versa* M1 A0 [1]
- 10 large / strong (constant) magnetic field B1
nuclei rotate about direction of field / precess (1)
radio frequency / r.f. pulse B1
causes resonance in nuclei , nuclei absorb energy (1)
(pulse) is at the Larmor frequency (1)
on relaxation / nuclei de-excite emit (pulse of) r.f. B1
detected and processed B1
non-uniform field (superimposed) B1
allows for position of nuclei to be determined B1
and for location of detection to be changed (1)
(B6 plus any two extra details, 1 each, max 2) B2 [8]
- 11 (a) (i) frequency of carrier wave varies M1
in synchrony with displacement of information signal A1 [2]
- (ii) 1. zero (accept constant) B1 [1]
2. upper limit 530 kHz B1
lower limit 470 kHz B1
changes upper limit → lower limit → upper limit at 8000 s^{-1} B1 [3]
- (b) e.g. more radio stations required / shorter range
more complex electronics
larger bandwidth required
(any two sensible suggestions, 1 each) B2 [2]
- 12 (a) (i) picking up of signal in one cable M1
from a second (nearby) cable A1 [2]
- (ii) random (unwanted) signal / power B1
that masks / added to / interferes with / distorts transmitted signal B1 [2]
(allow this mark in (i) or (ii))
- (b) if P is power at receiver,
 $30 = 10\lg(P / (6.5 \times 10^{-6}))$ C1
 $P = 6.5 \times 10^{-3} \text{ W}$ C1
loss along cable = $10\lg(\{26 \times 10^{-3}\} / \{6.5 \times 10^{-3}\})$ C1
= 6.0 dB C1
length = $6.0 / 0.2 = 30 \text{ km}$ A1 [5]